1. Identity

Document	STAR-PKG-STAR-J2-EN · v1.3 · 2026-10-16
Package	star-j2
Version	0.1.0
Capability	star.orbit.j2-secular
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_j2

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_j2-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_j2-0.1.0-py3-none-any.whl	3304813347d0e74e84b67052047c589f32208b37903ec5e663e87bc14445e9cf
star_j2.py	e97f9628cbc4ebb6cbfcf5c74143e1bc5a7dce2ba68e3192b2a4c259b82d460f
star_j2-0.1.0-source.zip	59707a979b4dff33934e35f81c06f8373466812329ce7b9fd094db1914d9ac81

5. What it does, and its limits

First-order secular effect of J2 on mean elements: node regression, perigee rotation, mean-anomaly rate, nodal period, sun-synchronous inclination; measured within 0.4 % of Orekit's Eckstein-Hechler theory and of a numerical integration.

Mean rates only: no short-period terms, no J2-squared, J3 or higher harmonics, no drag, no third bodies. Not a propagator. For mission design at the 0.5 % level; a station-keeping budget needs a full theory or an integration.

- `raan_rate_deg_day`, `argp_rate_deg_day`, `mean_anomaly_rate_deg_day`, `nodal_period_s` for mean elements (a in km, e, inclination in degrees), EGM96/WGS-84 constants by default, overridable by keyword.
- `sun_synchronous_inclination_deg(a_km, e=0)`: the inclination whose node advances 360 degrees per tropical year; `CRITICAL_INCLINATION_DEG` = 63.4349...: the perigee does not rotate.
- Accuracy, measured on 51 orbits: node rate within 0.31 % of Orekit's Eckstein-Hechler theory (J2 only, mean elements) and within 0.38 % of a numerical integration of the equations of motion with J2; perigee rate within 0.013 deg/day of the numerical integration. The formulas are first order in J2: differences of this size are the model, not a rounding error.
- Every function returns a finite float or raises `ValueError`: non-numeric, boolean, NaN or infinite input; a <= 0; e outside [0, 1); inclination outside [0, 180]; perigee inside the Earth; non-positive mu, re or j2; an orbit too high to be sun-synchronous (above about 12 350 km semi-major axis for a circular orbit).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_j2_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-j2}: What the Earth's oblateness does to an orbit, on average},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_j2/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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